

AAYUSH PANDA

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TECHNICAL SKILLS

Programming Languages: C, C++, Rust, Python, Java, TypeScript, SQL, Haskell, Scheme, Bash, HTML/CSS, x86 assembly
Machine Learning: Transformers, RNNs, Diffusion models, VAEs, PyTorch, Numpy
Databases: SQL dbs, Firestore, Neo4J, Chroma

EDUCATION

University of Waterloo

Bachelor of Computer Science, Honours Co-op

GPA: 3.92/4.00

Sep. 2024 – Apr. 2028

EXPERIENCE

ML & NLP Engineering Intern

Sep. 2025 – Apr. 2026

Huawei Canada

- Developed algorithms for KG generation from natural language documents, with named entity linking
- Improved performance of Neo4j knowledge graph updates using Merkle trees. Reduced latency by 95%
- Developed document indexing for AI text editor that achieved SOTA performance for extremely large documents (100+ pages)
- Adapted a hybrid ASR model architecture to work with natural language context to guide transcription, reduced WER by ~ 5%
- Developed real-time multimodal assistant agent for HarmonyOS, with 95% accuracy on visual tasks like slideshow editing.

Laboratory Assistant

Apr. 2025 – Dec. 2025

University of Waterloo Multi-Sensory Brain and Cognition Lab (MBC)

- Research on how batsmen's eyes track baseballs, measuring visual perception bandwidth, and correlating with batting ability
- Developed hybrid/synthetic data pipelines to train a **U-net** model to detect baseballs in captured video
- Developed correspondence algorithm to synchronise data from high speed head mounted camera and eye tracking data

Undergraduate Research Assistant

Apr. 2025 – Aug. 2025

University of Waterloo Data Privacy Lab

- Research on private record linkage protocols using locality sensitive hashing, supervised by Dr. Florian Kerschbaum
- Developed a differentially private method of estimating max histogram bin size, used for a novel private set intersection method
- Developed k-d cuckoo hashing algorithm that reduced expected-time complexity of PSI algorithm by an order of magnitude

Software Engineering Intern (Contract)

Mar. 2025 – Apr. 2025

Toma (YC W25)

- Developed automated lead qualification service that evaluates dealerships by calling them with a conversational voice AI.
- Developed unified call dashboard with synchronised audio, text transcripts and interactive server logs

PROJECTS

🔗 **Concord** — Rust, Vulkan, CUDA, HLSL

- Cross-platform runtime in Rust for accelerated computation, with a unified ISA and memory model for CPU/GPU operations.
- Architected bytecode ISA and VM runtime with native cooperative multitasking and FFI
- Developed highly modular compilation toolchain with inline bytecode support to allow extensions for elegant native extensions
- Created CLI interactive debugger with VM state inspection and variable watching

🔗 **Semantify** — Python, React, FastAPI, Sentence Transformers, UMAP, Ollama

- Built a system that semantically organises **1000+ files** into a directory structure in seconds using embedding models and custom hierarchical clustering and subtree merging algorithms
- Developed an interactive visualization of document embeddings to allow intuitive semantic exploration
- Implemented a RAG-powered chat interface to intelligently query uploaded documents with source citation

🔗 **Phased Array SONAR** — C++, Python, Xtensa LX6 microprocessor, AVR RISC processors

- Engineered a sub-degree precision beam-steering phased SONAR array with real-time radar-style display on oscilloscopes.
- Designed a waveguide to reduce inter-element pitch and suppress grating lobes, enhancing beam directivity.
- Built a phased array simulator suite to visualize beamforming, steering, and focus behaviors in 2D.

PATENTS

CA 3222437: Device for redirection of optical beams using virtual gratings generated by stationary waves.

CA 3119717: Compliant mechanism for operating flight control surfaces of a remotely piloted aircraft.

AWARDS

University of Waterloo President's Research Award

Jane Street Estimathon @ UWaterloo (2024): First place

Hack the North 2022: Winner (out of 829 participants)

PicoCTF: 2nd place in Canada, 14th (top 0.001%) globally